

PROJECT OVERVIEW

# LocalPDF Prototype

LocalPDF is a privacy-first browser toolkit for common document operations. The first milestone focuses on merging PDF files without sending them to a server.

## Problem

Online PDF tools may require uploads, accounts, or intrusive advertising.

## Audience

Students, freelancers, teachers, and small teams.

## Value

Simple local processing with clear privacy labels.

## Milestone

A reliable merge workflow with tests and cancellation.

USER FLOW

# Merge PDF Journey

| Step | User action   | System response                |
|------|---------------|--------------------------------|
| 1    | Select files  | Validate type and size         |
| 2    | Reorder files | Update the final sequence      |
| 3    | Start merge   | Show progress and allow cancel |
| 4    | Download      | Provide the merged document    |

The flow must remain usable with keyboard navigation and on a 320-pixel-wide screen.

TECHNICAL PLAN

## Architecture

The browser reads files with the File API, processes them in a Web Worker, creates the output as a Blob, and exposes a temporary object URL for download.

Upload system - 100%



Validation - 90%



Merge engine - 75%



Automated tests - 60%



No document content should be sent to analytics or stored in localStorage.

## QUALITY PLAN

## Definition of Done

| Check           | Expected                   |
|-----------------|----------------------------|
| Page count      | Sum of all input pages     |
| Order           | Matches the visible list   |
| Encrypted input | Human-readable error       |
| Large input     | No frozen interface        |
| Cancel          | Stops processing safely    |
| Output          | Opens in major PDF viewers |

*A feature is complete only when output correctness, error handling, documentation, and tests are all present.*